

Análisis Estadístico de Datos
Examen Argumentativo

Nombre: _____ Matrícula: _____

Instrucciones: Muestra cómo obtienes tu respuesta. Puedes usar Excel o Minitab para tus cálculos de probabilidades o percentiles, pero no uses otros archivos o hagas uso de la IA. Respeta el código de ética del curso y del Tecnológico de Monterrey.

1. A quality-control inspector examines 10 products. Each product has a probability of 0.30 of being defective, independently of the others.

Let X denote the number of defective products.

What is the probability that **exactly 4 products are defective**?

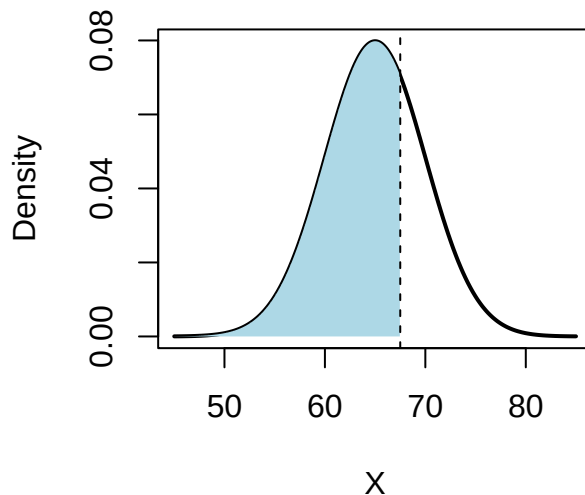
- (a) 0.3504
 - (b) 0.2001
 - (c) 0.1029
 - (d) 0.8497
2. A random sample of size $n = 49$ was obtained from a population. The sample mean was $\bar{x} = 87$ and the sample standard deviation was $s = 11$.

Compute the standard error of the sample mean using

$$SE(\bar{x}) = \frac{s}{\sqrt{n}}.$$

Round your answer to two decimal places.

3. The following figure shows a normal distribution with mean $\mu = 65$ and standard deviation $\sigma = 5$. The shaded region represents $P(X \leq 67.5)$ for a normally distributed random variable X .



Compute the probability represented by the shaded area. Round your answer to four decimal places.